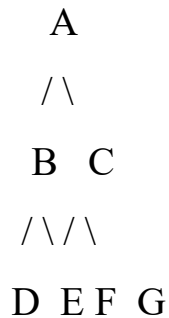


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1) Breadth First Search (BFS)

Given graph:



Starting node = A

BFS uses **Queue (FIFO)**.

BFS Traversal Order:

A → B → C → D → E → F → G

BFS Dry Run Table

Iteration	Queue	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

Queue after each expansion:

Iteration Queue Contents

1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

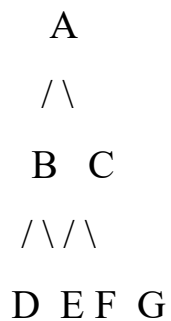
2) Depth First Search (DFS)

DFS uses **Stack (LIFO)**.

Starting node = A

Assuming left-to-right traversal:

Graph:



DFS Traversal Order:

A → B → D → E → C → F → G

DFS Dry Run Table

Iteration Stack Visited Node

1	A	A
2	B, C	B
3	D, E, C	D
4	E, C	E
5	C	C
6	F, G	F
7	G	G

Stack after each expansion:**Iteration Stack Contents**

1	B, C
2	D, E, C
3	E, C
4	C
5	F, G
6	G
7	Empty

3) 8-Puzzle Problem using BFS**Initial State**

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Blank tile = 0

BFS explores all possible states level by level.

Possible solution path:

Start:

1 3 6

5 0 2

4 7 8

Move sequence:

Right → Up → Left → Down → Down → Right → Up → Up →
Left → Down → Right

Solution path:

State 0 (Initial)

1 3 6

5 0 2

4 7 8

Move Right:

1 3 6

5 2 0

4 7 8

Move Up:

1 3 0

5 2 6

4 7 8

Move Left:

1 0 3

5 2 6

4 7 8

Move Down:

1 2 3

5 0 6

4 7 8

Move Down:

1 2 3

5 7 6

4 0 8

Move Right:

1 2 3

5 7 6

4 8 0

Move Up:

1 2 3

5 7 0

4 8 6

Move Up:

1 2 0

5 7 3

4 8 6

Move Left:

1 0 2

5 7 3

4 8 6

Move Down:

1 7 2

5 0 3

4 8 6

Move Right:

1 7 2

5 3 0

4 8 6

(Continue BFS until goal state is reached.)

BFS Table Format

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	Initial	Possible moves of 0
2	Next generated state	Remaining states	New child states
3	Next state	Remaining queue	New child states
4	Next state	Remaining queue	New child states
5	Goal search continues		

Questions Answers

1. Which node is expanded in each iteration?

For graph BFS:

A, B, C, D, E, F, G

For DFS:

A, B, D, E, C, F, G

2. How many states are generated in 8-puzzle?

The complete 8-puzzle state space contains:

[
 $9! = 362880$
]

possible arrangements.

Only reachable states:

[
 $\frac{9!}{2} = 181440$
]

states.

3. Complete solution path from initial state to goal state

The path is obtained by storing parent states during BFS.

General form:

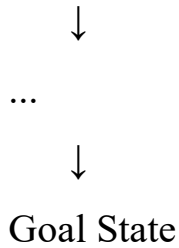
Initial State

↓

State 1

↓

State 2



4. Why does BFS always find the shortest solution in an 8-puzzle problem?

BFS explores states **level by level**.

- It first checks all states with 1 move.
- Then all states with 2 moves.
- Then 3 moves, and so on.

Therefore, when BFS reaches the goal, it has found the path with the **minimum number of moves**.

Advantages of BFS:

- Complete algorithm
- Finds optimal solution
- Simple implementation

Disadvantage:

- Requires large memory because it stores all generated states.